Artur Iasenovets

Information Security Research Engineer

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> **Research Area of Interest:** Cryptography; Formal Methods and Verification
> **Prospective Supervisors:** Prof. Benedikt Bünz (Courant); Prof. Siyao Guo (Shanghai)

Research Interests

privacy-enhancing technologies, private information retrieval, private pattern search, searchable encryption, homomorphic encryption, blockchain privacy.

Education

Chongqing University of Posts and Telecommunications, Chongqing, China
◇ **M.S.** in Network and Information Security — Sep. 2024–Jul. 2027 *(expected)*
◇ Supervisor: Prof. Tang Fei
◇ Average mark: 95/100

Ukhta State Technical University, Ukhta, Russia
◇ **B.S.** in Information Systems and Technologies — Mar. 2022–Jun. 2024
◇ GPA: 4.45/5.00

St. Petersburg State University of Telecommunications, St. Petersburg, Russia
◇ **B.S. studies** in Information Systems and Technologies — Sep. 2019–Mar. 2022

Selected Papers

Artur Iasenovets, Fei Tang, H. Zhu, P. Wang, and L. Liu. "Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval," arXiv:2511.02656, 2025. Manuscript under review.

H. Zhu, F. Tang, W. Qin, J. Shan, P. Wang, and Artur Iasenovets. "EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption," *Journal of King Saud University – Computer and Information Sciences*, 2025.

Artur Iasenovets and Fei Tang. "MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework," *International Research Journal*, no. 1 (163), 2026.

Piyumi M. S. Rajapaksha*, Artur Iasenovets*, Yinxue Yi, and Fei Tang. "TWAPPA-FRL: Trust-Weighted Privacy-Preserving Aggregation for Federated Reinforcement Learning in Collaborative Intrusion Detection," 2026. Manuscript under review. *Equal contribution.

Artur Iasenovets and Fei Tang. "CBPIR: Content-Based Private Information Retrieval over Structured Databases via Homomorphic Encryption," accepted at NSS-SocialSec 2026.

Research Experience

Graduate Researcher @ CQUPT — Sep. 2024–Present

- **HLF-CPIR:** Designed and implemented BGV-based private information retrieval in Go using Lattigo for private world-state queries in Hyperledger Fabric, with query privacy under an HBC peer model based on the IND-CPA security of BGV.
- **TWAPPA-FRL:** Designed the architecture and implemented CKKS-based private aggregation of DQN model updates using TenSEAL, with IND-CPA-based client-update privacy under an HBC aggregator model and explicit leakage analysis.
- **EVMK-SSE:** Analyzed the threat model with an HBC data owner and user and a malicious cloud server; reviewed the query-privacy argument in the real/ideal model, based on OT security and the pseudorandomness of CI-rOPRF/PRG under explicitly defined search- and access-pattern leakage.
- **CBPIR:** Designed and implemented a Rust prototype using OpenFHE, combining CKKS-based FPS/RFPS private pattern search, client-side match-to-record mapping, and BGV-based windowed CPIR for private multi-record retrieval, with explicit leakage and correctness analysis under an HBC server model.

Awards and Achievements

Chinese Government Scholarship (CSC) for Graduate Studies 2024–2027
Full scholarship for the entire M.S. programme at CQUPT.

Sber Student Accelerator Regional Demo Day 2023
Completed the Sber Student Accelerator with the AcademAI team and participated in the North-Western District regional demo day.

Winner, EdTech Project Competition 2023
Presented LLM-based educational platform and won the competition organized by the MIPT Phystech School, the Fund for Development of Phystech Schools, and Startech.Base.

Enhanced State Academic Scholarship 2023
Awarded by USTU for outstanding academic excellence, with the enhanced amount based on documented achievements in the university's index-based student rating system.

MIPT & Sber Phystech Gigachat Challenge 2023
Participated in the Phystech Gigachat Challenge Hackathon, developing an LLM-based educational platform using GigaChat and Kandinsky APIs within a 48-hour sprint.

Kaspersky Cyberimmunity Hackathon 2.0 2023
Participated in Kaspersky's Cyberimmunity Hackathon in Fall 2023, placing 11th out of 100+ teams for developing traffic monitoring solutions for low-altitude airspace security.

First Degree Diploma, Severgeoecotech 2022
Awarded first place in the Modern Information Technologies section of the 23rd International Youth Scientific Conference.

Supporting certificates

Industry Internships

Information Security Engineer @ Positive Technologies Aug. 2024–Feb. 2025

- Developed Python tools with unit-test coverage for validating Windows paths, network configurations, and endpoint-control policies.
- Built an automated malware-processing pipeline with an Airflow DAG that collected samples from MalwareBazaar, stored artifacts in S3, and scanned them with YARA.
- Analyzed EVTX logs, PowerShell, and malicious VBA macros; extracted IoCs and mapped TTPs to MITRE ATT&CK for CVE-2020-5902 and CVE-2023-38831.

Other Experience

Software Engineer @ AcademAI Sep. 2023–Aug. 2024

- Developed the application layer with Next.js/TypeScript application, Python/Flask services, PostgreSQL/Prisma data layer, and Auth.js authentication for an LLM-based educational platform used by computer science university faculty and students.
- Integrated LangChain-based generation workflows with AsyncIO and Redis for asynchronous LLM orchestration and intermediate state management.
- Containerized the application with Docker, configured Nginx, deployed to cloud-hosted VPS infrastructure, and built the GitHub Actions CI/CD workflow.

Junior Software Engineer @ ZVEK "Progress" LLC May. 2022–Aug. 2023

- Designed and developed a full-stack warehouse QR-inventory system using Python, FastAPI, and MySQL, replacing a manual tracking process entirely.
- Supported approximately 1,000 records and 20+ concurrent users; added unit-test coverage, Git-based development, debugging, and Linux deployment.

Skills

- **Programming:** Go, Python, Rust, TypeScript, SQL, Bash, LaTeX
- **Systems and infrastructure:** Linux, Proxmox VE, VMware, Docker, Git, CI/CD, Nginx, Ansible, Airflow, S3
- **Cryptographic libraries:** Lattigo, SEAL, TenSEAL, OpenFHE
- **Cryptography:** Lattice-based, symmetric-key, public-key
- **Security engineering:** YARA, malware analysis (PS/VBA), IoC/TTP analysis, MITRE ATT&CK
- **Languages:** Russian — native; English — C1; Chinese — HSK3